Introduction: Evidence Gaps in a Rapidly Expanding Field

The Free Energy Principle and Active Inference Framework

The Free Energy Principle (FEP), introduced by Karl Friston, proposes that self-organizing systems maintain their structural and functional integrity by minimizing variational free energy—an upper bound on sensory surprise friston2006free, friston2010free. Under this principle, living systems are cast as approximate Bayesian inference engines that build generative models of their environment and act to reduce the discrepancy between predicted and observed states. Active Inference (AIF) extends this picture from passive perception to goal-directed behavior: agents select actions that bring about observations consistent with their preferred states, unifying perception, learning, and decision-making within a single variational framework parr2022active, friston2017active. Since its initial formulation for sensorimotor control, AIF has been applied to navigation, visual foraging, language comprehension, social cognition, and multi-agent coordination. Bayesian mechanics sakthivadivel2023bayesian has further strengthened the mathematical foundations of the FEP by grounding Markov blanket dynamics in the physics of belief-based

Challenges Posed by Rapid Literature Growth

The active inference literature has grown at a compound annual rate of 16.99% across 2005–2026, with annual output accelerating sharply after 2013. While early research concentrated on theoretical neuroscience, the field has since diversified across biology (C5), robotics (C2), computational psychiatry (C4), algorithm scaling (B), and formal mathematics (A1). This multi-disciplinary expansion creates three interrelated challenges. First, tracking which core theoretical claims—such as FEP universality or the physical realism of Markov blankets—are well-supported, contested, or merely assumed becomes increasingly difficult as the corpus grows. Second, because the relationship between mathematical formalisms and empirical evidence is frequently implicit, systematic evidence synthesis demands substantial manual effort. Third, new entrants must navigate a literature weighted toward broad qualitative philosophy (A2), interspersed with specialized applied subfields.

Traditional narrative reviews attempt to address these challenges but are static, subjective, and quickly outdated. Systematic

Related Work and Prior Meta-Analyses

Several prior efforts have surveyed aspects of the Active Inference landscape. Sajid et al. sajid2021active compare active inference with alternative decision-making frameworks; Da Costa et al. dacosta2020active synthesize the discrete-state-space formulation; Lanillos et al. lanillos2021active survey robotics applications; Smith et al. smith2021computational provide a tutorial bridging theory and empirical data; and Millidge et al. millidge2021understanding examine information-theoretic foundations of exploration behavior. Ramstead et al. ramstead2018answering extend the FEP to questions of biological self-organization, while Pezzulo et al. pezzulo2015active connect active inference to homeostatic regulation. Millidge millidge2024retrospective provides a practitioner's retrospective confirming that AIF's strongest demonstrated results arise from novel discrete generative models, while scalability relative to deep reinforcement learning remains the field's central open challenge.

Closest to our work, Knight, Cordes, and Friedman knight2022fep conducted a systematic literature analysis of publications using the

Synergizing Knowledge Graphs and LLMs

Recent systematic literature initiatives underscore a powerful reciprocal synergy between Large Language Models (LLMs) and Knowledge Graphs: LLMs parse unstructured text to rapidly extract semantic claims, efficiently populating the structured, queryable architecture of the graph quevedo2025combining, li2024unifying. We adopt the *nanopublication* groth2010anatomy—a minimal, machine-readable unit of scientific evidence comprising a core assertion bound to explicit provenance metadata—as the fundamental serialization format for this extracted knowledge.

This Study: Approach and Overview

This paper presents a computational meta-analysis of the Active Inference literature ($N = 849$). Rather than relying exclusively on bibliometric metadata or slow manual coding, we deploy a Large Language Model (LLM) to "read" each paper's abstract and assess its relationship to eight core hypotheses within the FEP paradigm. We serialize these assessments as nanopublications—each encoding an assertion ("Paper X supports Hypothesis Y") coupled with the LLM's natural-language reasoning and confidence score. The resulting knowledge graph aggregates these nanopublications and links them to paper metadata, citation networks, subfield classifications, and hypothesis definitions. A citation-weighted scoring formula quantifies the net evidence for or against each hypothesis, producing scores in $[-1, 1]$ that reflect both the direction and strength of published evidence.

Research Questions

This meta-analysis addresses four primary research questions:

1. **RQ1 (Field Structure):** What is the disciplinary structure and growth trajectory of the Active Inference literature, and how are papers distributed across the three domains—Core Theory (A), Tools & Translation (B), and Application Domains (C)?
2. **RQ2 (Growth Dynamics):** What are the temporal growth dynamics of the field, and which subfields are experiencing the most rapid expansion?
3. **RQ3 (Hypothesis Evidence):** What is the current balance of evidence for and against the eight standard hypotheses, and how has this balance evolved over time? (See hypothesis dashboard and assertion figures in the hypothesis results.)
4. **RQ4 (Tooling Readiness):** What is the state of software tooling and infrastructure for Active Inference research, and what gaps remain?

Scope and Delimitations

This study focuses on the English-language peer-reviewed and preprint literature retrievable from arXiv, Semantic Scholar, and OpenAlex. Our search scope begins at 2005—chosen to capture Energy-Based Model and variational Bayesian antecedents (Helmholtz machines, VAEs, early Bayesian brain formulations dayan1995helmholtz, lecun2006tutorial) that share deep mathematical foundations with variational free energy minimization and predated the Free Energy Principle label introduced in 2006 friston2006free. The scope includes both the core Active Inference and Free Energy Principle literature and adjacent Energy-Based Model research where it intersects with variational inference or generative modeling—capturing the growing convergence between these traditions. We do not include book chapters or monographs not indexed by these sources, software documentation, or non-English publications. Domain classification uses keyword matching rather than expert annotation—a deliberate trade-off favoring reproducibility over precision, whose consequences we quantify in the field overview.

Principal Contributions

This work makes five contributions:

1. **A multi-source retrieval and deduplication pipeline** for Active Inference literature, using a canonical identifier hierarchy across three academic databases.

2. **A nanopublication-based knowledge graph schema** encoding directed, confidence-scored assertions about eight core hypotheses with full provenance tracking.

3. **A quantitative field overview** characterizing the growth, domain distribution (A/B/C taxonomy), citation topology, and latent topic structure of the Active Inference literature. The field's computational maturity is underscored by recent benchmark results: AXIOM heins2025axiom demonstrates AIF agents learning Gameworld 10k in minutes using object-centric world models, while Friston et al. friston2025active introduce Renormalization Generative Models (RGMs) that achieve 99.8% MNIST accuracy with 90% less data, pointing toward scalable multi-agent architectures. Collective AIF has been